ANNUAL REVIEWS

Annual Review of Sociology

What Makes Weak Ties Strong?

Minjae Kim[1] and Roberto M. Fernandez[2]

[1]Department of Management—Organizational Behavior, Jones Graduate School of Business, and Department of Sociology, Rice University, Houston, Texas, USA; email: minjaekim@rice.edu

[2]Department of Management and Department of Organization Studies, Sloan School of Management, Massachusetts Institute of Technology, Cambridge, Massachusetts, USA; email: robertof@mit.edu

ANNUAL REVIEWS CONNECT
www.annualreviews.org
• Download figures
• Navigate cited references
• Keyword search
• Explore related articles
• Share via email or social media

ws.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

Annu. Rev. Sociol. 2023. 49:177–93

First published as a Review in Advance on March 21, 2023

The *Annual Review of Sociology* is online at soc.annualreviews.org

https://doi.org/10.1146/annurev-soc-030921-034152

Copyright © 2023 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.

OPEN ACCESS CC BY

Keywords

strength of weak ties, social networks, diffusion, structure versus agency, bridging ties, computational social science

Abstract

We raise two challenges concerning the validity of arguments underlying Granovetter's strength of weak ties (SWT) thesis: (*a*) whether weak ties are actually bridges, i.e., they help reach more socially distant actors than strong ties, and (*b*) whether weak ties transmit information effectively enough so that weak ties' alleged structural properties make them more useful than strong ties. In the course of reviewing subsequent research that has made progress in addressing these challenges, we identify both potential limits and possibilities for the SWT thesis. We argue for the importance of identifying how actors' agency—i.e., the way people use their ties—may affect social networks' value. We conclude by summarizing some outstanding questions that progress on the SWT thesis has generated.

INTRODUCTION

Social network analysis is an important sociological export to not only other social sciences (e.g., Gee et al. 2017a) but also physical sciences and engineering (e.g., Onnela et al. 2007). Among the sociological giants who have advanced the field is Mark Granovetter, who has written two of the three most-cited sociology articles according to the Web of Science as of this writing. We focus this review in celebration of the fiftieth anniversary of arguably his most influential article, "The Strength of Weak Ties" (Granovetter 1973). The article's elegance lies in its two main theoretical moves that may now be taken for granted in social network analysis. One is that it focused on ties as the unit of analysis (see also Coleman et al. 1966). By doing so, Granovetter sought to identify conditions under which actors could benefit from certain types of ties, above and beyond when some groups of connected actors might be advantaged (e.g., Milgram 1967, Boissevain 1968, Mitchell 1969; cf. Coleman 1988). Another is that the article linked a tie's strength—understood as interpersonal closeness—to a structural property, specifically, the extent to which the tie bridges gaps in the social structure (see also Harary et al. 1965, Liu & Duff 1972). Based on these key moves, the article used job search in the labor market [see Fernandez 2021 for a recent review on the strength of weak ties (SWT) in this domain], community structure, and social movements (Granovetter 1978, Granovetter & Soong 1983) to illustrate how the bridging property of weak ties can also be an important insight in information diffusion (see also Granovetter 1995).

There are many reasons to celebrate these theoretical innovations. But when one considers how much confidence we have in the validity of the celebrated article's arguments alone—as opposed to how generative it has been for research in various disciplines—the celebration might be a bit quieter. The major goal of this review is to interrogate the empirical validity of the SWT thesis and thus to provide a surer theoretical edifice for one of our discipline's most celebrated theories. Toward this end, we identify two major sets of challenges and discuss the progress that has been made in addressing them in the 50 years since Granovetter's landmark article first appeared. The first issue we discuss is the degree to which tie strength is correlated with the structural property of bridging social worlds. The second is related to questions about how strong and weak ties work in the diffusion of information. By analyzing these two challenges, and the progress that has been made to address these challenges, we probe the limits and possibilities of the SWT thesis. This exercise ultimately suggests that underlying the SWT thesis may be an "overstructured" view of human action (cf. Granovetter 1985) and that the strength of weak ties coming from their structural property as bridges should be considered against agency with which people use their social networks. While the structural property proposed in the SWT argument may be a necessary condition for diffusion of information, it may not be a sufficient condition. We conclude with suggestions for future research that can continue advancing knowledge on these key theoretical challenges.

LOOKING UNDER THE HOOD OF THE STRENGTH OF WEAK TIES

We begin by focusing on two key drivers behind the SWT thesis. The first is that a tie's strength is inversely related to social distance. That is, weak ties tend to be bridges, reaching socially distant information that is nonredundant with information that the focal individual directly has access to via strong ties (see also Liu & Duff 1972). The logic underlying this claim is illustrated in the original article in the discussion of the "forbidden triad"—if two individuals are each connected to the focal individual via strong ties, the two individuals must also share a strong tie (cf. Heider 1946, Davis 1963, Holland & Leinhardt 1971). The flip side of this logic is that if two individuals are each connected to the focal individual but the two individuals are not connected to one another (i.e., an "open triad"), the ties that the two individuals have with the focal individual must be weak yet must

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

be the only pathways through which the two individuals can reach one another. Consequently, weak ties likely help reach nonredundant information more than strong ties. This is one of the logical steps made to derive the counterintuitive implication that people are more likely to get jobs through weak ties than strong ties. The null hypothesis, or the implicit intuition before the SWT thesis, was that people are more likely to get jobs through strong ties (e.g., family, close friends) because close contacts are more likely to help and/or know relevant job information. Contrary to this intuition, the SWT thesis focused on the benefit of receiving nonredundant information via weak ties.

If the first driver behind the SWT is one of the explicit contributions of Granovetter's (1973) article, the second driver behind the SWT is more implicit: that weak ties transmit information unproblematically, i.e., to the extent that a tie's effectiveness as a conduit of information is not so low as to cancel the advantage coming from its structural property. Note that the strength behind weak ties in Granovetter's theory specifically relies on weak ties' ability to reach useful information, and not on other features, such as ties' ability to exert influence on behalf of the job seeker and/or moral support that contacts connected via weak ties can provide to the job seeker for their presence alone (cf. Lin et al. 1981, Krackhardt 1992, Bian 1997, Tian & Lin 2016). In sum, while these arguments by no means suggest that strong ties are useless (Granovetter 1983), they identify the counterintuitive benefit of weak ties. Yet, even though the veracity of these two claims is critical to the overall argument underlying the SWT, neither the original article nor later work, with a few important exceptions, directly addressed these issues. Therefore, in the rest of this review, we focus on progress that has been made on this front, the limits and possibilities of Granovetter's original argument identified by this progress, and remaining tasks for future research to tackle.

CHALLENGE I: DOES WEAK NECESSARILY MEAN BRIDGING?

Our first challenge is whether or not weak ties are necessarily bridges. One perspective on this challenge is that the use of a tie's strength as an indicator of its structural bridging property was not analytically necessary, but it was a convenient solution to empirical challenges in discovering a tie's structural property. That is, the SWT thesis boils down to how a tie's structural property as a bridge—that it reaches a social circle with which the focal individual does not come into contact often—helps obtain useful information, and not that a weak tie is more likely to have such a structural property. The claim that weak ties are more likely to be bridging thus may seem neither necessary nor critical to the underlying contribution of the SWT thesis. In fact, this is one of advances that Burt (1992) made in *Structural Holes*, as the network benefit was theorized to depend solely on the focal individual's structural position and not tie strength (see also Reagans & McEvily 2003). This perspective may get even more purchase if we think that tie strength was used merely as a proxy for researchers in the 1970s (when Granovetter's 1973 article was published) in measuring a tie's structural property. However, with the advanced infrastructure and technology that have now enabled scholars to gather fine-grained data on ties' structural property (e.g., cheaper survey instruments or access to proprietary network data such as email exchanges, as in Kleinbaum 2012), using tie strength as an indicator of a structural property may no longer be necessary.

Reflecting this notion, Rajkumar et al. (2022), in their article titled "A Causal Test of the Strength of Weak Ties," conceptually conflate tie strength and structural property in testing the SWT thesis (see also Gee et al. 2017b). Using LinkedIn data, they experimentally manipulate to whom the focal individual is recommended to connect—those with whom the focal individual shares mutual friends versus those with whom the focal individual does not share any. Via the direct manipulation, they provide evidence of a causal effect of having bridging versus bonding

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

ties. Specifically, they find that (*a*) the focal individual is more likely to get a job via those with whom the focal individual does not share many mutual friends, but that (*b*) this higher likelihood of getting jobs based on sharing few mutual friends persists only to a point, after which there were diminishing marginal returns. In short, moderate bridges are most helpful for job searches, compared with less or extremely bridging ties. Nevertheless, these findings leave the effect of having weak ties compared with strong ties ambiguous. That is, their experimental manipulation concerns the bridging structural property of ties and not tie strength, the latter of which is measured (via interaction intensity) but not manipulated in their study. Insofar as there are off-diagonal situations—i.e., ties that are weak and bonding, or strong and bridging (cf. Brashears & Quintane 2018, Neal 2022)—characterizing weak ties as necessarily bridging ties and strong ties as necessarily bonding ties may be problematic.

To be sure, one may reasonably argue that the core logic underlying the SWT thesis addresses the advantage based on having bridging ties; thus, studying the effect of bridging (rather than weak) ties is more productive. But if this is the objective, such a test—as that of Rajkumar et al. (2022)—should instead be referred to as a causal test of bridging ties, or of occupying structural holes (Burt 1992). Although this is still extremely valuable to network scholars, it is not a direct test of the SWT thesis. And for the sake of conceptual precision, there are at least two additional reasons why it is important to treat tie strength as an independent conceptual construct and empirically investigate the correlation between tie strength and structural property. The first reason is that the original SWT thesis's innovation was partly based on its discovery of weak ties' potential value, which might have been overlooked as less useful in social networks than strong ties. Insofar as the intuitive null hypothesis was based on tie strength per se, the analytical move that connected tie weakness to bridging might conflate these two features, potentially confounding our understanding of how useful each type of tie is. Thus, examining the relationship between tie strength and structural property would be an important exercise in theoretical elaboration. But there is a second, more practical reason as well. Although it now may be easier for scholars to measure ties' structural properties, assessing ties' structural properties may still be difficult for social actors. For instance, Krackhardt (1987, 1988, 1990, 1992) has powerfully identified accurate knowledge of social networks in and of itself as an important driver of social advantage. Of course, it may also be difficult to have accurate perceptions of tie strength and structural property of one's own and/or others' social networks. But being able to deduce a tie's structural property based on its strength can help social actors triangulate the value of their and others' social networks.

Hence, empirically verifying the inverse relationship between tie strength and social distance becomes critical in assessing the validity of the SWT thesis. While Friedkin (1980) examined the multidisciplinary network of biological scientists and provided evidence of the predicted inverse relationship, the focus on biological scientists—even though multidisciplinary—inevitably limited the extent to which bridges can be examined. After all, bridges, strong or weak, may be observable only among very socially disparate groups (e.g., biological scientists and graphic artists) and when they are examined together in relatively global networks. Furthermore, there are theoretical reasons to think that strong ties may also connect socially distant individuals. Consider multiplex ties (e.g., Fischer 1982, McPherson et al. 2001), where two individuals may have different types of relationships with one another (e.g., professional and personal; Luo & Chung 2005, Ingram & Zou 2008). For instance, siblings may have a strong personal tie even though they may have little overlap in their professional activities and contacts. Consequently, two individuals connected via a strong tie may still help one another reach information that is socially distant.

Such multiplex ties may have practical as well as substantive implications. On one hand, it makes measuring tie strength trickier than it already is. While Granovetter (1973, p. 1361) defined tie

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

strength as "a combination of the amount of time, the emotional intensity, the intimacy (mutual confiding), and the reciprocal services which characterize a tie," this definition did not give a clear guidance as to which of these factors should be prioritized if they are not highly correlated with one another (e.g., if two colleagues work and spend most of the day together professionally but are not emotionally close). Marsden & Campbell (1984) find that there are several distinct aspects of tie strength—e.g., time spent in a relationship and the depth of the relationship—but they are not necessarily highly correlated with one another (see also Brashears & Quintane 2018). Further complicating the matter, Bearman & Parigi (2004, p. 535) find that, while talking about "important matters" in the General Social Survey may seem indicative of a strong tie, "the connection between characteristics of discussion networks and achievement of instrumental ends—for example, getting a job or enhancing social support—is tenuous" (cf. Brashears 2011). This ambiguity may reflect the fact that ties can be strong for different reasons: Tie strength may represent a wide range of characteristics, from a commitment to serve one another to a sense of ingroup or feelings of emotional closeness (Krackhardt 2003).

On the other hand, the lack of consensus on the proper measure of tie strength might not be that problematic. For instance, Li & Piezunka (2020) recently identified the possibilities of "role transitions," where people who are connected in one domain (e.g., the personal sphere) may also play an important role in another domain (e.g., the professional sphere; see also Portes & Sensenbrenner 1993, Uzzi 1997). This suggests that the more important empirical feature is that there be some variance in the degree to which a tie is weak or strong, however it is defined. If a role transition is possible, having close emotional attachments may translate to a high sense of ingroup and/or a higher willingness to provide professional help. This, of course, is not to assume that every role transition in multiplex ties is possible, as some professional relationships are explicitly or implicitly banned from becoming strong personal relationships (e.g., universities often prohibit an academic advisor and a graduate student from being romantically involved). But the larger point is that even though there may indeed be multiple dimensions to social relationships, focusing on one or a few that may translate to the overall strength of a tie may be sufficient to represent the underlying likelihood that a tie is used when the need arises (cf. Marsden & Campbell 2012).

In this context, Park et al. (2018) empirically test the claim that weak ties are more likely to be bridges that help reach more socially distant information. They examine 11 culturally diverse population-scale networks on four continents—encompassing 56 million Twitter users and 58 million mobile phone subscribers. In their Twitter data, they posit that a social tie between two individuals takes place when each sends at least one @mention or @reply tweet to the other. Beyond this low threshold, they posit that the more frequently such bidirected communications happen, the stronger the tie is. A bidirected phone call (as opposed to a missed call) constitutes a tie in the mobile phone data, with more frequent calls representing stronger ties. Their measure of social distance, or "tie range," as they call it, is the number of intermediary ties required to reach from an individual node to its neighbor if their direct tie were removed. Given these measures of tie strength and social distance, they find that there is a U-shaped relationship (with tie range on the x axis and tie strength on the y axis): While weak ties tend to reach more distance than some strong ties embedded within a small circle of contacts, extremely long-range ties are nearly as strong as those strong ties embedded in a small circle. This evidence may constitute the most direct test on one of the most important claims made in the SWT thesis, i.e., that weaker ties reach more socially distant actors. Their findings suggest an amendment to the original SWT thesis, implying that the tie strength's relationship with socially distant information is not linear, but is curvilinear and U-shaped. Insofar as it is a tie's structural property in reaching

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

nonredundant information that drives the advantage associated with weak ties (cf. Burt 1992), it seems more useful to identify the strong ties that help reach such information. Intuitively, it may even be easy to identify such strong ties—e.g., high school friends who stay close and get together during holidays but do not have much overlap in professional contacts. This proposition, of course, is also subject to empirical verification.

Another question raised by the findings of Park et al. (2018) is where "bridgingness" comes from. That is, Granovetter's logic for why weak ties tend to be bridges was based on the forbidden triad, where two people connected to the focal individual via strong ties are unlikely to lack a strong tie between themselves. Therefore, for the focal individual to be a connector between the two people, the ties that connect the focal individual and the two people have to be weak. The flip side of this logic naturally implies that bridges—which tend to be weak—are more likely to decay (Burt 2002, Kleinbaum 2018). However, the findings of Park et al. (2018) imply not only that the forbidden triad may not necessarily be binding but also that the high strength of bridges may sustain such ties over time. To the former, Jahani et al. (2022) suggest that people with "previous disruptive life events," such as migration to another state or attending college outside of the home state (in the United States) tend to have more bridges that are strong as well. While suggestive, this between-person analysis does not address the question fully, in that each individual (as a within-person characteristic) likely has ties that are both strong and long, as found by Park et al. The origin of such bridges thus remains fertile for further investigations.

At the same time, note that the overarching claim behind the SWT thesis may incorporate these findings from Park et al. (2018) unproblematically. In particular, the following two statements may simultaneously be true: (*a*) some strong ties help reach extremely distant information, but (*b*) the most helpful information comes from medium-distant ties, which are also still more likely to be weaker than not-so-distant ties. Consider how some extremely distant job information may not be useful: For example, the job seeker works in the IT industry, and she receives, via a far-reaching strong tie, information about a job in the construction industry. Extremely far-reaching strong ties may be able to help reach such extremely distant and nonredundant information, but for the purposes of a job search, medium-distant information may still be the most relevant and thus helpful, and weak ties may be the most likely to help reach it (cf. Obukhova 2012). In fact, Rajkumar et al. (2022) find that information gathered from medium-distant ties is more helpful than that from extremely far-reaching ties or short-distance ties. If it is the case that such most-helpful, medium-distant ties in the findings by Rajkumar et al. (2022) indeed correspond to Park et al.'s (2018) weakest ties, the overarching claim behind the SWT—that people are more likely to get a job through a weak tie than a strong tie—may still hold. This would be despite the theoretical amendment to the claim that weak ties on average help reach more distant information than (all) strong ties.

Another caveat for scholars of weak ties is that the findings of Park et al. (2018) may be reflective of the contemporary world where direct communications are more convenient than ever. In other words, the claim that weak ties help reach more distant information may have empirically been true in the 1970s when the SWT thesis was written, before the advent of cell phone technology and the Internet. To carry the example, friends from high school who regularly talk and get together during holidays but have little overlap in professional circles—i.e., a possible case of strong ties that help reach extremely distant information—may not have been able to maintain their ties as strongly if they could not talk via cell phone and/or communicate online between holidays. Of course, conversely, the (curvilinear, U-shaped) relationship between tie strength and distance, as found by Park et al., may have held true in the 1970s, but given that the extremely long strong ties

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

that Park et al. show are likely observable only in population-wide datasets such as Twitter and cell phone data, this may not have been visible in Granovetter's research design that examined right-truncated data (see also Fernandez 2021). Nevertheless, it remains plausible that the relationship between tie strength and the social distance a tie helps reach has evolved over time in human societies, and our opportunity to empirically verify this relationship may be capturing only one snapshot of a certain time period.

An upshot from work since the original SWT thesis examined the relationship between tie strength and social distance is that people may have strong ties that help them reach extremely distant information, while the usefulness of such distant information reached via strong ties is still to be tested (cf. Rajkumar et al. 2022). This lesson is certainly meaningful for sociologists who track progress in our understanding of the SWT thesis. But it also has practical implications: Even though a tie's structural property—i.e., whether a tie is a bridging tie as opposed to a bonding tie—seems to drive much of a tie's value as a conduit of information (although see the next section) and seems possibly proxied by a tie's strength, strong ties may be either. The key would be to identify which strong ties help reach nonredundant information and which help reach redundant information.

CHALLENGE II: DO WEAK TIES UNPROBLEMATICALLY RELAY INFORMATION?

Notwithstanding findings from Park et al. (2018), the SWT thesis remains viable to the extent that weak ties reach more distant information than some (or most) strong ties, and/or insofar as weak ties may reach exactly the distance where information is most valuable. Yet, these possibilities are still viable only if weak ties actually relay information, as far as information diffusion is concerned. That is, the fact that weak ties help connect two individuals with nonredundant information is a necessary condition for the SWT thesis, but it is not a sufficient condition unless weak ties also relay such information between distant social worlds.

This claim—that weak ties not only connect but also relay nonredundant information between actors in distant social worlds—may be implicit, but it is critical for work that has built on the SWT thesis to theorize how diffusion of information occurs. For instance, consider the line of research on the "small-world phenomenon," building on work by Watts & Strogatz (1998) and Watts (1999). Through simulations, they argue that information diffusion is slow when networks are highly clustered, but information diffusion becomes dramatically more efficient via "rewiring" only a few bonding ties into bridging ties that connect different social circles that are otherwise unconnected. This research has been hugely influential, with the earlier Watts & Strogatz (1998) article arguably as influential as the original SWT article and often referred to as a testament to the durability of the SWT thesis (see also Onnela et al. 2007). This argument also presumes that bridging ties are effective at relaying information, at least to some extent: If the likelihood of information transfer is miniscule in (presumably weak and) bridging ties, the fact that (presumably weak and) bridging ties connect two different social circles would be moot, as information would not travel regardless of the presence of such ties (e.g., see simulation results from Neal 2022).

In fact, subsequent work has proposed several reasons why information diffusion via weak ties—which may be more likely to constitute bridging ties—may not be as effective as Watts & Strogatz (1998) suggest. The first reason is that information transfer via social ties may not be automatic, and worse yet, information transfer via weak ties may be disproportionately more difficult than that via strong ties precisely because of the weak ties' bridging nature. The basic analogy that

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

Watts and Strogatz initially propose is that of disease contagion, where a social tie between two individuals would automatically and blindly exchange information. Such an analogy may be useful, especially when COVID-19 pushed us to apply social networks to our understanding of how quickly a group of individuals (e.g., college students) may be infected with a contagious disease in their institutional setting (e.g., Weeden & Cornwell 2020, Weeden et al. 2021). Yet, actors may not necessarily transfer information as automatically, especially when they hold monopolies over the information coming from one distant social circle to another. Weak and bridging ties provide value to brokers who possess them, precisely because brokers are able to control the use of such ties and extract profit from facilitating information transmission, exercising their "control benefit" in their powerful broker role (Burt 1992, Reagans & Zuckerman 2008).

The second reason why weak and bridging ties may not necessarily make information diffusion as efficient as Watts & Strogatz (1998) presume is that there may be a "diversity-bandwidth trade-off" (Aral & Van Alstyne 2011) where strong and bonding ties are able to transfer more complex information. Some empirical work has documented this trade-off, without always explicitly using this terminology (Szulanski 1996; Hansen 1999, 2002; Wu et al. 2008; Aral 2016; Bruggeman 2016; but see Brashears & Quintane 2018).

The third reason why weak ties may prevent information diffusion is because information transferred via weak ties may be less relevant than presumed to the potential beneficiary of that information. For an analogy, consider a line of work that has focused on the comparison between information achieved via other means (e.g., job information via formal means) and that achieved via social networks and suggested that the latter is more relevant to the information seeker based on homophilous networks (Mouw 2003, McDonald 2011, Cingano & Rosolia 2012, Horváth 2014). The same issue may apply to the comparison between information achieved via weak ties and that via strong ties, where the latter is more relevant for the information seeker (Obukhova 2012). As mentioned earlier, extremely distant information may not necessarily be as helpful. Two actors from completely different social worlds may have little overlap in not only their information but also their preferences, endowment, and knowledge. In this case, even though weak and bridging ties connect the two social worlds, the information possessed by each actor may not be that useful to the other.

Yet, the first (broker-interest-based resistance to information diffusion via weak ties) and third reasons (information via weak ties as less relevant) why weak and bridging ties may not necessarily make information diffusion dramatically more efficient are based on the bridging nature of weak ties. Brokers may withhold information transferred via weak ties not because their bridging ties are weak, but because their bridging ties help them reach nonredundant information and they are trying to profit from that nonredundant information. Also, the second reason (the diversity-bandwidth trade-off) suggests that information transferred via weak ties and that transferred via strong ties is likely to be qualitatively different (e.g., simple versus complex). However, McAdam & Paulsen (1993) hinted that information transmission may increase as a function of only tie strength in and of itself, all other factors being equal, by discovering that social movement activists were more likely to receive recruitment information via strong than weak ties. Their empirical innovation was not to select on the dependent variable—e.g., those who were recruited—but to study the entire pool of potential recruits and examine the drivers behind the recruitment. This then raises a possibility that tie strength in and of itself may be a determinant of information transmission, especially in risky environments such as social movement activism and illicit activities (cf. Baker & Faulkner 1993, 2003; Aven 2015).

Subsequently, two complementary lines of work have emerged to account for why strong ties, instead of weak ties, may drive more efficient information diffusion. One line of work focuses on the fact that people may come across more strong ties than weak ties in their daily lives. In

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

particular, Centola & Macy (2007) propose a model of "complex contagion," where information diffusion inherently requires multiple exposures to the same information, and thus strong ties that provide redundant information play a critical role. Part of their insight is that they flip weak ties' likelihood to transmit nonredundant information—i.e., the characteristic of weak ties that used to be considered their "strength"—into their weakness when it comes to the diffusion of "costly, risky, or controversial" information. In the situation of participating in a protest, for example, individuals have to come across the same information (e.g., how many people are willing to participate in a protest) multiple times in order to be reassured of its reliability. This line of work consequently implies that for certain types of information, weak and long ties might not be as useful for diffusion.

Nevertheless, the idea of complex contagion still assumes that weak ties transmit information as effectively as strong ties. That is, the "weakness of (weak and) long ties" hinges on the fact that a weak tie provides nonredundant information and not on the fact that a weak tie by itself acts differently from a strong tie in transferring information. By contrast, another line of work identifies strong ties as a driver of more efficient information diffusion by focusing on how one strong tie may transmit information more effectively than one weak tie. One example comes from Kim & Fernandez (2017), who directly show through a vignette experiment using US-based MBAs that people are more willing to relay job information to a friend than an acquaintance. This line of work suggests that people would be more motivated to share more (fine-grained) information with people with whom they feel close (cf. Uzzi 1997). The reasons behind strong ties' higher likelihood to transfer information may vary, but a prominent one is higher motivation associated with strong ties: People are more motivated to help people they consider close to them over those with whom they share only weak ties (e.g., Griswold 1994). While it may seem intuitive that people may be more motivated to relay information via strong ties, this line of work provides the first direct evidence that strong ties are disproportionately more likely to transfer information than weak ties.

This is not to say that the motivation to transfer information is always higher for strong ties. In fact, Bond et al. (2018) find that people may be more willing to relay job information to (*a*) a stranger who is well qualified for the job or (*b*) a stranger when a referral bonus (as much as or more than $250) is paid, relative to (*c*) a friend who is less qualified and when no referral bonus is paid. In other words, the tendency for people to protect their reputation by referring only contacts who are qualified for the job (Smith 2005) may work against the higher likelihood of relaying information via strong ties. Also, weak ties may be able to provide benefits associated with their structural property (i.e., bridgingness) if they contain a high level of trust in actors facilitating information transfer (Levin & Cross 2004; cf. Aven et al. 2021, Iorio 2022). (Although if weak ties, on average, contain a lower level of trust between two parties, such tendencies for weak ties to transfer information reliably may be the exception rather than the rule.) Furthermore, the motivation to relay information may be even lower for some strong ties under some conditions. For instance, Marin (2012) documents that people may sometimes refrain from giving job information to their close contacts, fearing that such an action might be interpreted as implying that those contacts need better jobs. When one receives some information via a weak tie, one may think that the sender of the information is oblivious enough about one's circumstances that the sender is not passing judgments on one's preferences or circumstances. But there may not be the benefit of the doubt for two actors connected via a strong tie.

Furthermore, even if weak ties are less effective at relaying information than strong ties, weak ties may still be critical in making information diffusion more efficient. Insofar as weak ties help an actor reach a social distance that most strong ties are not able to help reach, weak ties' lower likelihood of information transmission may not be as problematic. That is, in some circumstances,

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

the nonredundant information that weak and long ties are able to reach may be so valuable that even if those ties are not very effective at relaying such information, the snippets of that information may still provide more value than strong ties that are much more effective at relaying useless information. This question of when information provided by weak ties may be valuable enough to "save" the SWT—despite their lower likelihood of information transfer—is an important topic for future research.

STRENGTH OF WEAK TIES AS AN OVERSTRUCTURED VIEW OF NETWORKS

The aforementioned work building directly or indirectly on the SWT thesis identifies when and why the structural advantage implied by weak ties may not necessarily stand. To summarize, this was done in two ways: First, the structural distance covered by weak ties may not be as long as that covered by some strong ties, and second, the structural distance covered by weak ties—even if long enough—may not give much information advantage when the weak ties do not relay information effectively enough.

Consequently, one major takeaway may be that the SWT thesis offers an overstructured view of networks. Perhaps ironically, this is analogous to an argument by another of Granovetter's (1985) influential articles, "Economic Action and Social Structure: The Problem of Embeddedness." In this article, his argument built on a critique by Wrong (1961) that sociology was conceiving of individuals as "oversocialized" actors who would blindly internalize some norms and conventions governing their communities (cf. Centola et al. 2005, Willer et al. 2009, Kim et al. 2022; for a recent review, see Guhin et al. 2021). In that paper, rather than relying on some amorphous forces driving socialization, Granovetter identified networks as drivers of people's actions as well as conduits of socialization. Analogously, in this review, we identify the focus on weak ties' structure as the overdeterministic driver of information diffusion. Progress made to address the two aforementioned challenges—and especially the second challenge—instead has revealed that information diffusion may depend on not only social networks' structure but also the agency with which social actors use their social networks.

The debate between structure and agency as the more important determinant of various social outcomes may be as old as the field itself. But the focus in the rest of this review is more narrowly to point out a context in which information diffusion may be hampered due to actors' agency, even when the same structural conditions (e.g., presence and range of networks) are present. Specifically, one such condition may be when the use of social networks is perceived to be instrumental. In his memorable aphorism, Burt (1992, p. 72) put it as: "Judging friends for efficiency is an interpersonal flatulence from which friends will flee." Despite the "relentless encroachment of the market" (Kuttner 1997; see also Zelizer 1978, Almeling 2007), using a personal relationship for instrumental purposes is often perceived as morally objectionable. Therefore, one persistent challenge for social actors is how to mute the perception that they are approaching one another with instrumental motives, even if their motives truly are instrumental.

To this point, Schilke & Rossman (2018) (see also Rossman 2014), who supply experimental evidence, and Child (2021), who supplies qualitative evidence, suggest that people "obfuscate" their instrumental motives in order to mitigate their exchange partners' moral offense. This does not necessarily mean that productive use of social networks requires there be no perceptions of instrumental motives. Rather, the argument by Schilke & Rossman (2018) and Child (2021) is precisely that the strategies they identify are used to obfuscate the transactional motives and not to necessarily remove the transactional motives. For instance, junior scholars may reach out to senior scholars in academic conferences in order to get their work noticed and receive feedback

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

(and not because the junior scholars are big fans and thus think that they would intrinsically enjoy speaking with the senior scholars), and senior scholars may be very aware of such instrumental motives. Yet, with the obfuscatory strategies, junior scholars may be able to construct "veneers and underlayments" (Winship 2004) with which they can not only plausibly deny instrumental motives behind the social relationships but also help their exchange partners have some plausible deniability of knowing of such instrumental motives. This is analogous to one interpretation of the situation in the "emperor's dilemma" right before the child yells out that the emperor is naked. Even though exchange partners are aware of one another's instrumental motives, obfuscatory strategies help them pretend as if they do not have such motives and/or are not aware of one another's instrumental motives, successfully carrying out the transaction and maintaining the social decorum often required to preserve the social relationship.

Naturally, the SWT may also depend in part on the extent to which weak ties (versus strong ties) help people obfuscate their instrumental motives and, perhaps more importantly, the extent to which weak ties allow people to maintain plausible deniability of such motives. Consider a situation where one's employer offers handsome compensation for referring a successful hire. In this case, one may hesitate to contact distant acquaintances (i.e., people connected to one via weak ties) due to the fear that contacting them after a long pause in communication would invite suspicion of self-interested motives. Instead, one may turn to people with whom communication is more frequent (i.e., people connected via strong ties) in order to make the referral appear more natural and freer of such ulterior motives. This example likely extends beyond the labor market context as well. For instance, when a person contacts a distant acquaintance after a long pause for any purpose, the receiver of information may raise questions about what prompted them to break the pause in the first place: The sender must have some extrinsic motive behind relaying the information!

At the same time, this problem around the appearance of instrumentality when communicating via weak ties may be less salient when plausible deniability is broken. Recall the situation in the emperor's dilemma. Once the child yells out that the emperor is naked, it should not be possible for people to convincingly hide their awareness of the emperor's nakedness (unless they can plausibly pretend not to have heard it; cf. Chwe 2003, Adut 2005). The analogy to situations of information transfer via networks would be when distant acquaintances are somehow made aware that the communication via weak ties after a long pause was based on instrumental motives. In such situations, Chan (2009) documents that weak ties tend to be more effective at realizing instrumental purposes. Furthermore, the cost of broken plausible deniability may be lower for weak ties than strong ties (cf. Marin 2012): One weak tie may not have as much emotional or even instrumental value as one strong tie (cf. Gee et al. 2017b). In other words, if a person's best friend appears to be using them for their own benefit, the person may break off the friendship and experience emotional distress, whereas the same person may not care much if a distant acquaintance seems to be using them. In that case, insofar as the plausible deniability around one's instrumental motives is broken, one may actually be more likely to transfer information via weak ties than strong ties. To be clear, these conjectures need to be empirically tested as well as theoretically established in a more systematic way. But they raise the potential significance of agency in further investigating weak ties' ability to facilitate information diffusion.

CONCLUSIONS AND FUTURE RESEARCH DIRECTIONS

The objective of this review is not to cover all the relevant literature published on the SWT thesis since its theoretical inception. Rather, it is to focus on two specific arguments made in the SWT thesis, assess progress made to test their validity, and raise further reasons why the SWT thesis may

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

be robust as well as questioned. As proud as we may be of the cross-disciplinary influence of the SWT, the validity of the underlying arguments and assumptions should be subject to empirical and theoretical inquiries, which we now can address better with the benefit of hindsight and advanced theoretical and empirical tools. The two challenges that this review raises—and the progress that has been made in addressing those challenges—should not be seen as diminishing the substantive generative value of the SWT thesis. Rather, the effort that has gone into identifying the potential limits and possibilities of the SWT by subsequent research should be understood as indicators of the lasting value of the insights offered by the SWT. Ultimately, this exercise raises several important questions for future research that may help probe limits and possibilities of the SWT thesis:

- Even though some strong ties help reach the most distant information (Park et al. 2018), are there situations where people find information received via (medium-range) weak ties more helpful than that received via (longer-range) strong ties (cf. Rajkumar et al. 2022)? If so, when and why? Addressing these questions would help us understand whether weak ties still provide more benefit than strong ties despite their shorter range than that of some strong ties.
- Are people better at recognizing the strength of their and others' network ties than they are at observing the structural properties (e.g., range) of those networks? Addressing this question would help assess the value of studying tie strength as a proxy for structural properties.
- Relatedly, while Park et al. (2018) argue that some strong ties are farther-reaching than weak ties, can people identify such strong ties? Addressing this question would further understanding of whether and when people can identify the likeliest sources of nonredundant information based on tie strength alone.
- Are there (technological or other) conditions under which people can maintain long-range ties as strong? Addressing this question would help identify the conditions under which tie strength and structural properties are correlated—inversely, as the original SWT thesis implies—or otherwise.
- Has the U-shaped relationship between tie strength and social distance changed over time (due to technological or other reasons)? Or has that relationship remained the same, but Granovetter was examining a right-truncated dataset compared with Park et al. (2018)?
- Where does a tie's bridgingness come from? When and why can people build bridges? Addressing these questions may help uncover the origins of structural connectedness between people of different backgrounds and/or preferences in a way that does not reinforce existing segregation, norms, or inequality (Blau & Schwartz 1984; cf. Centola 2015).
- What are the contexts in which the lower likelihood of information transfer via weak ties ultimately makes long-range weak ties less valuable than short-range strong ties? In turn, what are the contexts in which longer-range weak ties still trump short-range strong ties in value, despite the lower likelihood of information transfer via weak ties? Addressing these questions would help identify when and why structural properties are more important than agency implied by tie strength.
- Under what conditions can people deny their ulterior (e.g., instrumental) motives when communicating via weak ties through which communication has been infrequent? In turn, under what conditions and for what types of ties do people unproblematically embrace their exchange partners' ulterior motives that fail to be obfuscated? Addressing these questions would help us assess the extent to which people can overcome (and are limited by) the moral obligations implied by weak versus strong ties.

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

Addressing these questions would help further triangulate the value of weak ties and serve to complement other extant work that tests the potential limits of the SWT thesis. While space limitations prevent us from reviewing these works, we want to conclude by directing the reader to three additional issues that this review has not focused on. First, as suggested by Mouw (2003, 2006), social ties may be correlates of other social characteristics that drive network advantage and, relatedly, weak ties may provide benefit over strong ties not necessarily because weak ties have causal effects but because weak ties are correlates of other social characteristics that do provide social advantage. Progress has been made here by sociologists and others, who now often use (field) experiments and other identification strategies to isolate such causal effects of social ties (e.g., person fixed effects in Obukhova & Lan 2013, Greenberg & Fernandez 2016; missed opportunities to network due to weather-dependent flight cancellation in Kneeland & Kleinbaum 2021; and the aforementioned Rajkumar et al. 2022). But future work may also investigate different conditions under which (strong versus weak) ties function as causal forces behind value creation/capture, as some ties may bring value only under certain circumstances.

The second issue is understanding the marginal effect of weak versus strong ties. This issue has been raised by Fernandez (2021), who identifies Gee et al. (2017a) as setting the frontier by finding, using Facebook data, that a single strong tie is more helpful for getting a job than a single weak tie, while a collection of weak ties is more helpful than a collection of strong ties. This is consistent with findings from Yakubovich (2005).

Finally, while this review explicitly does not focus on gender, race, and other demographic characteristics of people connected via weak versus strong ties, answers to the above questions may differ widely based on them. For instance, insofar as networks are indeed homophilous based on gender and/or race (McPherson et al. 2001), it may be the case that weak ties for women and racial minorities—which might connect them to powerful individuals who tend to be white and male and to whom women and minorities may not be connected via strong ties (cf. McPherson & Smith-Lovin 1982, Moore 1990)—might be more beneficial than they are for white men (cf. Ibarra 1992, Hagan 1998, Lin 2000, Kossinets & Watts 2009, Abraham 2020). Relatedly, Obukhova & Kleinbaum (2022) recently documented that women engage in more "scouting," where they try to find employers and career options that give them a fair chance at professional success, while engaging in as much "schmoozing" as men, where they reach out to at least as many potential contacts. This might imply that women are more likely to build ties—and perhaps more weak and long-range ties—in order to maintain options that give them equal opportunities. However, at the same time, Brands & Kilduff (2014) argue that women who occupy the same brokerage position as men are perceived to be less warm, implying a disadvantage for women who may possess more weak ties (see also Brands & Mehra 2019). Future research along these lines thus should investigate the questions raised above separately for men and women, whites versus racial minorities, and so on. This imperative stands especially when, depending on the question, the fact that the same answers apply for men versus women, or white versus racial minorities, may still be an important theoretical and empirical discovery vis-à-vis the SWT thesis, but it might also serve to advance understanding of the structural basis of gender and race inequality.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

We are grateful to Patrick Park for helpful comments on an earlier draft of this manuscript.

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

LITERATURE CITED

Abraham M. 2020. Gender-role incongruity and audience-based gender bias: an examination of networking among entrepreneurs. *Adm. Sci. Q.* 65(1):151–80

Adut A. 2005. A theory of scandal: Victorians, homosexuality, and the fall of Oscar Wilde. *Am. J. Sociol.* 111(1):213–48

Almeling R. 2007. Selling genes, selling gender: egg agencies, sperm banks, and the medical market in genetic material. *Am. Sociol. Rev.* 72(3):319–40

Aral S. 2016. The future of weak ties. *Am. J. Sociol.* 121(6):1931–39

Aral S, Van Alstyne M. 2011. The diversity-bandwidth trade-off. *Am. J. Sociol.* 117(1):90–171

Aven B, Morse L, Iorio A. 2021. The valley of trust: the effect of relational strength on monitoring quality. *Organ. Behav. Hum. Decis. Process.* 166:179–93

Aven BL. 2015. The paradox of corrupt networks: an analysis of organizational crime at Enron. *Organ. Sci.* 26(4):980–96

Baker WE, Faulkner RR. 1993. The social organization of conspiracy: illegal networks in the heavy electrical equipment industry. *Am. Sociol. Rev.* 58(6):837–60

Baker WE, Faulkner RR. 2003. Diffusion of fraud: intermediate economic crime and investor dynamics. *Criminology* 41(4):1173–206

Bearman P, Parigi P. 2004. Cloning headless frogs and other important matters: conversation topics and network structure. *Soc. Forces* 83(2):535–57

Bian Y. 1997. Bringing strong ties back in: indirect ties, network bridges, and job searches in China. *Am. Sociol. Rev.* 62(3):366–85

Blau PM, Schwartz JE. 1984. *Crosscutting Social Circles: Testing a Macrostructural Theory of Intergroup Relations*. New York: Academic

Boissevain J. 1968. The place of non-groups in the social sciences. *Man* 3(4):542–56

Bond BM, Labuzova T, Fernandez RM. 2018. At the expense of quality. *Sociol. Sci.* 5:380–401

Brands RA, Kilduff M. 2014. Just like a woman? Effects of gender-biased perceptions of friendship network brokerage on attributions and performance. *Organ. Sci.* 25(5):1530–48

Brands RA, Mehra A. 2019. Gender, brokerage, and performance: a construal approach. *Acad. Manag. J.* 62(1):196–219

Brashears ME. 2011. Small networks and high isolation? A reexamination of American discussion networks. *Soc. Netw.* 33(4):331–41

Brashears ME, Quintane E. 2018. The weakness of tie strength. *Soc. Netw.* 55:104–15

Bruggeman J. 2016. The strength of varying tie strength: comment on Aral and Van Alstyne. *Am. J. Sociol.* 121(6):1919–30

Burt RS. 1992. *Structural Holes: The Social Structure of Competition*. Cambridge, MA: Harvard Univ. Press. 1st ed.

Burt RS. 2002. Bridge decay. *Soc. Netw*. 24(4):333–63

Centola D. 2015. The social origins of networks and diffusion. *Am. J. Sociol.* 120(5):1295–338

Centola D, Macy M. 2007. Complex contagions and the weakness of long ties. *Am. J. Sociol.* 113(3):702–34

Centola D, Willer R, Macy M. 2005. The emperor's dilemma: a computational model of self-enforcing norms. *Am. J. Sociol.* 110(4):1009–40

Chan CS-C. 2009. Invigorating the content in social embeddedness: an ethnography of life insurance transactions in China. *Am. J. Sociol.* 115(3):712–54

Child C. 2021. How to sell a friend: disinterest as relational work in direct sales. *Sociol. Sci.* 8:1–25

Chwe MS-Y. 2003. *Rational Ritual: Culture, Coordination, and Common Knowledge*. Princeton, NJ: Princeton Univ. Press

Cingano F, Rosolia A. 2012. People I know: job search and social networks. *J. Labor Econ.* 30(2):291–332

Coleman JS. 1988. Social capital in the creation of human capital. *Am. J. Sociol.* 94:S95–120

Coleman JS, Katz E, Menzel H. 1966. *Medical Innovation: A Diffusion Study*. Indianapolis, IN: Bobbs-Merrill. 1st ed.

Davis JA. 1963. Structural balance, mechanical solidarity, and interpersonal relations. *Am. J. Sociol.* 68(4):444–62

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

Fernandez RM. 2021. Strength in weak ties in the labor market: an assessment of the state of research. In *Personal Networks: Classic Readings and New Directions in Egocentric Analysis*, ed. B Pescosolido, EB Smith, pp. 251–64. Cambridge, UK: Cambridge Univ. Press
Fischer CS. 1982. *To Dwell Among Friends: Personal Networks in Town and City*. Chicago: Univ. Chicago Press
Friedkin N. 1980. A test of structural features of Granovetter's strength of weak ties theory. *Soc. Netw.* 2(4):411–22
Gee LK, Jones JJ, Burke M. 2017a. Social networks and labor markets: how strong ties relate to job finding on Facebook's social network. *J. Labor Econ.* 35(2):485–518
Gee LK, Jones JJ, Fariss CJ, Burke M, Fowler JH. 2017b. The paradox of weak ties in 55 countries. *J. Econ. Behav. Organ.* 133:362–72
Granovetter M. 1978. Threshold models of collective behavior. *Am. J. Sociol.* 83(6):1420–43
Granovetter M. 1983. The strength of weak ties: a network theory revisited. *Sociol. Theory* 1:201–33
Granovetter M. 1985. Economic action and social structure: the problem of embeddedness. *Am. J. Sociol.* 91(3):481–510
Granovetter M. 1995. *Getting a Job: A Study of Contacts and Careers*. Chicago: Univ. Chicago Press. 2nd ed.
Granovetter M, Soong R. 1983. Threshold models of diffusion and collective behavior. *J. Math. Sociol.* 9(3):165–79
Granovetter MS. 1973. The strength of weak ties. *Am. J. Sociol.* 78(6):1360–80
Greenberg J, Fernandez RM. 2016. The strength of weak ties in MBA job search: a within-person test. *Sociol. Sci.* 3:296–316
Griswold W. 1994. *Cultures and Societies in a Changing World*. Thousand Oaks, CA: SAGE. 4th ed.
Guhin J, Calarco JM, Miller-Idriss C. 2021. Whatever happened to socialization? *Annu. Rev. Sociol.* 47:109–29
Hagan JM. 1998. Social networks, gender, and immigrant incorporation: resources and constraints. *Am. Sociol. Rev.* 63(1):55–67
Hansen MT. 1999. The search-transfer problem: the role of weak ties in sharing knowledge across organization subunits. *Adm. Sci. Q.* 44(1):82–111
Hansen MT. 2002. Knowledge networks: explaining effective knowledge sharing in multiunit companies. *Organ. Sci.* 13(3):232–48
Harary F, Norman RZ, Cartwright D. 1965. *Structural Models: An Introduction to the Theory of Directed Graphs*. New York: Wiley. 1st ed.
Heider F. 1946. Attitudes and cognitive organization. *J. Psychol.* 21(1):107–12
Holland PW, Leinhardt S. 1971. Transitivity in structural models of small groups. *Comp. Group Stud.* 2(2):107–24
Horváth G. 2014. Occupational mismatch and social networks. *J. Econ. Behav. Organ.* 106:442–68
Ibarra H. 1992. Homophily and differential returns: sex differences in network structure and access in an advertising firm. *Adm. Sci. Q.* 37(3):422–47
Ingram P, Zou X. 2008. Business friendships. *Res. Organ. Behav.* 28:167–84
Iorio A. 2022. Brokers in disguise: the joint effect of actual brokerage and socially perceived brokerage on network advantage. *Adm. Sci. Q.* 67(3):769–820
Jahani E, Fraiberger SP, Bailey M, Eckles D. 2022. Origins and consequences of long ties in social networks. SocArXiv g2nkq. **https://doi.org/10.31235/osf.io/g2nkq**
Kim M, Fernandez RM. 2017. Strength matters: tie strength as a causal driver of networks' information benefits. *Soc. Sci. Res.* 65:268–81
Kim M, Zuckerman E, Hahl O, Poskanzer E. 2022. *When truth trumps facts: five studies on partisan moral flexibility in American politics*. Work. Pap., Rice Univ., Houston, TX
Kleinbaum AM. 2012. Organizational misfits and the origins of brokerage in intrafirm networks. *Adm. Sci. Q.* 57(3):407–52
Kleinbaum AM. 2018. Reorganization and tie decay choices. *Manag. Sci.* 64(5):2219–37
Kneeland MK, Kleinbaum AM. 2021. *On agency and its limits: the asymmetric effects of offsites on network tie formation*. Work. Pap. 3520640, Tuck Sch. Bus., Dartmouth Univ., Hanover, NH
Kossinets G, Watts DJ. 2009. Origins of homophily in an evolving social network. *Am. J. Sociol.* 115(2):405–50
Krackhardt D. 1987. Cognitive social structures. *Soc. Netw.* 9(2):109–34

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

Krackhardt D. 1988. Predicting with networks: nonparametric multiple regression analysis of dyadic data. *Soc. Netw.* 10(4):359–81
Krackhardt D. 1990. Assessing the political landscape: structure, cognition, and power in organizations. *Adm. Sci. Q.* 35(2):342–69
Krackhardt D. 1992. The strength of strong ties: the importance of philos in organizations. In *Networks and Organizations: Structure, Form, and Action*, ed. N Nohria, RG Eccles, pp. 216–39. Boston: Harvard Bus. Sch. Press
Krackhardt D. 2003. Constraints on the interactive organization as an ideal type. In *Networks in the Knowledge Economy*, ed. R Cross, A Parker, L Sasson, pp. 324–35. Oxford, UK: Oxford Univ. Press
Kuttner R. 1997. *Everything for Sale: The Virtues and Limits of Markets*. New York: Knopf
Levin DZ, Cross R. 2004. The strength of weak ties you can trust: the mediating role of trust in effective knowledge transfer. *Manag. Sci.* 50(11):1477–90
Li JB, Piezunka H. 2020. The uniplex third: enabling single-domain role transitions in multiplex relationships. *Adm. Sci. Q.* 65(2):314–58
Lin N. 2000. Inequality in social capital. *Contemp. Sociol.* 29(6):785–95
Lin N, Ensel WM, Vaughn JC. 1981. Social resources and strength of ties: structural factors in occupational status attainment. *Am. Sociol. Rev.* 46(4):393–405
Liu WT, Duff RW. 1972. The strength in weak ties. *Public Opin. Q.* 36(3):361–66
Luo X, Chung C-N. 2005. Keeping it all in the family: the role of particularistic relationships in business group performance during institutional transition. *Adm. Sci. Q.* 50(3):404–39
Marin A. 2012. Don't mention it: why people don't share job information, when they do, and why it matters. *Soc. Netw.* 34(2):181–92
Marsden PV, Campbell KE. 1984. Measuring tie strength. *Soc. Forces* 63(2):482–501
Marsden PV, Campbell KE. 2012. Reflections on conceptualizing and measuring tie strength. *Soc. Forces* 91(1):17–23
McAdam D, Paulsen R. 1993. Specifying the relationship between social ties and activism. *Am. J. Sociol.* 99(3):640–67
McDonald S. 2011. What you know or who you know? Occupation-specific work experience and job matching through social networks. *Soc. Sci. Res.* 40(6):1664–75
McPherson JM, Smith-Lovin L. 1982. Women and weak ties: differences by sex in the size of voluntary organizations. *Am. J. Sociol.* 87(4):883–904
McPherson M, Smith-Lovin L, Cook JM. 2001. Birds of a feather: homophily in social networks. *Annu. Rev. Sociol.* 27:415–44
Milgram S. 1967. The small world problem. *Psychol. Today* 2:60–67
Mitchell JC. 1969. *Social Networks in Urban Situations: Analyses of Personal Relationships in Central African Towns*. Manchester, UK: Manchester Univ. Press
Moore G. 1990. Structural determinants of men's and women's personal networks. *Am. Sociol. Rev.* 55(5):726–35
Mouw T. 2003. Social capital and finding a job: Do contacts matter? *Am. Sociol. Rev.* 68(6):868–98
Mouw T. 2006. Estimating the causal effect of social capital: a review of recent research. *Annu. Rev. Sociol.* 32:79–102
Neal ZP. 2022. Sometimes weak ties are just weak. *Science* 377(6612):1304–10
Obukhova E. 2012. Motivation versus relevance: using strong ties to find a job in urban China. *Soc. Sci. Res.* 41(3):570–80
Obukhova E, Kleinbaum AM. 2022. Scouting and schmoozing: a gender difference in networking during job search. *Acad. Manag. Discov.* 8(2):203–23
Obukhova E, Lan G. 2013. Do job seekers benefit from contacts? A direct test with contemporaneous searches. *Manag. Sci.* 59(10):2204–16
Onnela J-P, Saramäki J, Hyvönen J, Szabó G, Lazer D, et al. 2007. Structure and tie strengths in mobile communication networks. *PNAS* 104(18):7332–36
Park PS, Blumenstock JE, Macy MW. 2018. The strength of long-range ties in population-scale social networks. *Science* 362(6421):1410–13

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23

Portes A, Sensenbrenner J. 1993. Embeddedness and immigration: notes on the social determinants of economic action. *Am. J. Sociol.* 98(6):1320–50
Rajkumar K, Saint-Jacques G, Bojinov I, Brynjolfsson E, Aral S. 2022. A causal test of the strength of weak ties. *Science* 377(6612):1304–10
Reagans RE, McEvily B. 2003. Network structure and knowledge transfer: the effects of cohesion and range. *Adm. Sci. Q.* 48(2):240–67
Reagans RE, Zuckerman EW. 2008. Why knowledge does not equal power: the network redundancy trade-off. *Ind. Corp. Change* 17(5):903–44
Rossman G. 2014. Obfuscatory relational work and disreputable exchange. *Sociol. Theory* 32(1):43–63
Schilke O, Rossman G. 2018. It's only wrong if it's transactional: moral perceptions of obfuscated exchange. *Am. Sociol. Rev.* 83(6):1079–107
Smith SS. 2005. "Don't put my name on it": social capital activation and job-finding assistance among the Black urban poor. *Am. J. Sociol.* 111(1):1–57
Szulanski G. 1996. Exploring internal stickiness: impediments to the transfer of best practice within the firm. *Strateg. Manag. J.* 17(Suppl. 2):27–43
Tian FF, Lin N. 2016. Weak ties, strong ties, and job mobility in urban China: 1978–2008. *Soc. Netw.* 44:117–29
Uzzi B. 1997. Social structure and competition in interfirm networks: the paradox of embeddedness. *Adm. Sci. Q.* 42(1):35–67
Watts DJ. 1999. Networks, dynamics, and the small-world phenomenon. *Am. J. Sociol.* 105(2):493–527
Watts DJ, Strogatz SH. 1998. Collective dynamics of 'small-world' networks. *Nature* 393(6684):440–42
Weeden KA, Cornwell B. 2020. The small-world network of college classes: implications for epidemic spread on a university campus. *Sociol. Sci.* 7:222–41
Weeden KA, Cornwell B, Park B. 2021. Still a small world? University course enrollment networks before and during the COVID-19 pandemic. *Sociol. Sci.* 8:73–82
Willer R, Kuwabara K, Macy MW. 2009. The false enforcement of unpopular norms. *Am. J. Sociol.* 115(2):451–90
Winship C. 2004. Veneers and underlayments: critical moments and situational redefinition. *Negot. J.* 20(2):297–309
Wrong DH. 1961. The oversocialized conception of man in modern sociology. *Am. Sociol. Rev.* 26(2):183–93
Wu L, Waber B, Aral S, Brynjolfsson E, Pentland S. 2008. Mining face-to-face interaction networks using sociometric badges: predicting productivity in an IT configuration task. In *Proceedings of the International Conference on Information Systems* (*ICIS 2008*). Atlanta, GA: Assoc. Inf. Syst.
Yakubovich V. 2005. Weak ties, information, and influence: how workers find jobs in a local Russian labor market. *Am. Sociol. Rev.* 70(3):408–21
Zelizer VA. 1978. Human values and the market: the case of life insurance and death in 19th-century America. *Am. J. Sociol.* 84(3):591–610

Downloaded from www.annualreviews.org. Guest (guest) IP: 120.208.104.98 On: Wed, 28 Oct 2026 17:06:23